Scope: ICBHI 4-class respiratory sound classification, official 60/40 split. Part 1 (encoder comparison under the published protocols) is finished except for one DASS cell, Part 2 (proposed device-shift methods) has its cross-validation screening done and its final refit running. The refit tables are not in this report yet.

1. Setup

- **Data.** ICBHI 2017 (920 recordings, 6,898 cycles, 126 patients), official train/test split (4,142 / 2,756 cycles), 8 s cycles at 16 kHz. Classes: normal, crackle, wheeze, both. The 2-class score is a view of the same 4-class predictions (normal vs abnormal).
- **Metrics.** Sp = normal recall, Se = recall of abnormal cycles (correct class), Score = (Sp + Se) / 2, HS = harmonic mean of Sp and Se, per-class recall.
- **Training recipe** (Patch-Mix CL reference code): Adam, learning rate 5e-5, cosine schedule, batch 8 (DASS: 16, as in Lung-SRAD), 50 epochs, moving average 0.5, SpecAugment (160 frames, 48 bins), weight decay 1e-6. Patch-Mix CL: beta 1.0, tau 0.06 (DASS: 0.2), alpha 1.0.
- **Encoders.** AST, HTS-AT and DASS are trained in two modes (full fine-tuning, frozen encoder with trained head and projector); OPERA-CT, CLAP and HeAR only frozen. Each encoder gets its own native spectrogram (audited against the reference code; two mismatches found and fixed: HTS-AT now uses 10 s clips, DASS normalises each cycle). Patch-Mix is applied to the patch tokens of every encoder.
- **Protocol.** Seeds 0-4 (the papers use five seeds, no cross-validation). Numbers are mean ± SD over seeds at the epoch with the best test Score ("best on test", optimistic, as in the papers) and, in the last column, at epoch 50 (not selected on test).
- **Compute.** Two machines (RTX 5000 Ada and RTX 5090, 32 GB each); jobs run in parallel under a memory-gated scheduler and are resumable.

2. Part 1: encoder comparison

Encoder	Mode	Method	Seeds	Sp	Se	Score	HS	2-class Score	Score at epoch 50
AST	fine-tune	Patch-Mix CL	5	78.1±5.8	44.8±5.1	**61.4±1.3**	56.6±3.0	67.9±1.3	58.5±1.3
AST	fine-tune	Patch-Mix	5	79.2±5.2	39.8±5.1	**59.5±0.6**	52.7±3.5	66.0±1.9	56.6±0.8
HTS-AT	fine-tune	Patch-Mix CL	5	76.1±3.6	46.3±2.8	**61.2±0.5**	57.5±1.1	67.5±0.3	58.5±0.8
HTS-AT	fine-tune	Patch-Mix	5	77.4±3.8	44.9±2.8	**61.2±0.9**	56.7±1.3	68.2±0.8	59.1±0.6
DASS	fine-tune	Patch-Mix CL	5	71.8±3.8	49.2±2.0	**60.5±1.2**	58.3±0.9	67.1±1.3	57.9±0.9
DASS	fine-tune	Patch-Mix	5	75.6±3.6	47.5±2.5	**61.6±1.3**	58.3±1.4	67.6±1.0	59.7±1.2
AST	frozen	Patch-Mix CL	5	68.8±1.8	44.3±1.8	**56.6±0.3**	53.9±0.8	63.9±0.7	55.1±0.4
AST	frozen	Patch-Mix	5	76.7±3.4	37.9±2.7	**57.3±0.7**	50.7±1.9	62.9±0.9	56.5±0.5
HTS-AT	frozen	Patch-Mix CL	5	71.6±3.3	34.6±3.2	**53.1±0.3**	46.5±2.2	60.0±0.8	51.5±0.4
HTS-AT	frozen	Patch-Mix	5	79.8±4.7	28.5±6.0	**54.2±1.1**	41.5±6.0	59.9±3.1	50.1±0.4
DASS	frozen	Patch-Mix CL	5	69.6±2.5	40.8±2.4	**55.2±0.2**	51.3±1.2	61.5±0.6	54.1±0.2
DASS	frozen	Patch-Mix	4	75.4±3.4	36.7±3.4	**56.1±0.3**	49.2±2.4	60.9±0.2	54.5±0.1
CLAP	frozen	Patch-Mix CL	5	67.8±4.4	39.8±3.6	**53.8±0.5**	49.9±1.6	61.3±0.5	52.6±0.3

Encoder	Mode	Method	Seeds	Sp	Se	Score	HS	2-class Score	Score at epoch 50
CLAP	frozen	Patch-Mix	5	71.2±3.5	38.7±3.2	**54.9±0.4**	50.0±1.9	61.8±0.6	54.1±0.4
HeAR	frozen	Patch-Mix CL	5	87.5±8.2	20.0±8.2	**53.7±1.0**	31.6±10.5	56.0±1.9	51.1±0.2
HeAR	frozen	Patch-Mix	5	82.0±8.2	24.5±7.6	**53.3±0.8**	36.8±8.8	56.3±1.5	51.3±0.2
OPERA-CT	frozen	Patch-Mix CL	5	86.9±8.6	12.3±9.1	**49.6±0.3**	20.2±13.7	52.1±1.7	46.6±0.3
OPERA-CT	frozen	Patch-Mix	5	84.4±5.6	18.6±4.5	**51.5±0.7**	30.1±5.6	54.7±1.4	48.6±0.7

Per-class recall at the best-on-test epoch (mean ± SD, %):

Encoder	Mode	Method	Normal	Crackle	Wheeze	Both
AST	fine-tune	Patch-Mix CL	78.1±5.8	57.3±6.6	34.1±7.0	16.9±10.5
AST	fine-tune	Patch-Mix	79.2±5.2	50.7±3.8	30.6±6.1	15.4±14.7
HTS-AT	fine-tune	Patch-Mix CL	76.1±3.6	55.2±5.8	38.7±3.4	26.7±6.5
HTS-AT	fine-tune	Patch-Mix	77.4±3.8	54.7±3.0	35.6±4.8	25.3±10.6
DASS	fine-tune	Patch-Mix CL	71.8±3.8	52.6±2.1	53.7±4.2	21.4±3.5
DASS	fine-tune	Patch-Mix	75.6±3.6	51.3±3.4	48.7±2.5	26.9±4.4
AST	frozen	Patch-Mix CL	68.8±1.8	54.5±1.5	37.0±5.8	18.0±4.5
AST	frozen	Patch-Mix	76.7±3.4	53.8±3.2	19.7±4.5	15.0±4.7
HTS-AT	frozen	Patch-Mix CL	71.6±3.3	41.7±7.1	33.4±4.3	6.0±6.3
HTS-AT	frozen	Patch-Mix	79.8±4.7	38.0±11.1	22.6±4.2	1.0±1.1
DASS	frozen	Patch-Mix CL	69.6±2.5	45.3±3.3	42.0±3.2	16.9±3.6
DASS	frozen	Patch-Mix	75.4±3.4	43.6±2.1	34.8±8.2	10.1±5.1
CLAP	frozen	Patch-Mix CL	67.8±4.4	48.3±4.4	35.9±7.5	11.3±5.1
CLAP	frozen	Patch-Mix	71.2±3.5	48.1±4.5	35.0±2.6	5.7±2.1
HeAR	frozen	Patch-Mix CL	87.5±8.2	35.9±14.6	0.8±1.6	0.0±0.0
HeAR	frozen	Patch-Mix	82.0±8.2	39.5±10.2	7.5±6.8	2.2±2.3
OPERA-CT	frozen	Patch-Mix CL	86.9±8.6	20.5±16.3	2.3±1.4	1.5±1.3
OPERA-CT	frozen	Patch-Mix	84.4±5.6	33.1±7.8	0.9±0.8	0.3±0.6

Encoder	Mode	Method	Normal	Crackle	Wheeze	Both

Findings:

- **Fine-tuning.** AST, HTS-AT and DASS fall within about 1 point of each other (60.5-61.6, seed SD 0.5-1.3). The data do not rank them. Patch-Mix CL helps AST by about 2 points (61.4 vs 59.5) and does not help HTS-AT or DASS (DASS is 1.1 points higher with Patch-Mix alone).
- **Frozen encoders.** AST 56.6-57.3 > DASS 55.2-56.1 > CLAP 53.8-54.9, HTS-AT 53.1-54.2, HeAR 53.3-53.7 > OPERA-CT 49.6-51.5. Fine-tuning gains 4-8 points over freezing for the same encoder.
- **Class balance.** HeAR and OPERA-CT reach their Score through high Sp and very low Se (12-25) and almost never predict wheeze or both. AST, HTS-AT and DASS recover more abnormal cycles. DASS has the best wheeze recall (49-54) and the best Se (47-49) among fine-tuned encoders.
- **Epoch selection matters.** Best-on-test epochs are often 1-10, and Score at epoch 50 is about 3 points lower than the best-on-test value for fine-tuned runs. Published best-on-test numbers therefore overstate what a fixed training length achieves.

3. Comparison with the reference papers

Setting	Paper	Ours	Gap
AST, Patch-Mix only	59.46 ± 0.78 (Patch-Mix CL, ablation)	59.5 ± 0.6	0.0
AST, Patch-Mix CL	62.37 ± 0.61 (Sp 81.66, Se 43.07)	61.4 ± 1.3 (Sp 78.1, Se 44.8)	-1.0
AST, Patch-Mix CL, 2-class	68.71 ± 0.59	67.9 ± 1.3	-0.8
DASS, CE fine-tune	61.06 ± 1.27 (Lung-SRAD)	not run (we use Patch-Mix)	n/a
DASS, + regularization + AST-style Patch-Mix CL	62.42 ± 0.33	60.5 ± 1.2 (without regularization)	not like for like
DASS, Lung-SRAD (dual-axis mixing and blur)	64.48 ± 0.25	not run	n/a

The AST reproduction is credible (Patch-Mix only matches; Patch-Mix CL is about 1.4 standard errors below, inside the acceptance window of 1.0 point set beforehand). The DASS run is plain DASS and cannot be compared with the regularized variants. Published fine-tuned references for SG-SCL (61.71), LungAdapter (62.40), Audio-CLAP (62.56), BTS (63.54) and PAFA (64.84) have no matching run here. No paper reports frozen encoders (HTS-AT, HeAR, OPERA-CT, CLAP, DASS) on ICBHI 4-class, so those rows are our own measurements.

4. Part 2: proposed device-shift methods, cross-validation screening (AST, Patch-Mix CL)

Three-fold patient-grouped cross-validation on the training patients (the test set is not used). Each spectrum-correction (SC) arm was run with both coefficient bounds (dynamic, static); the better bound per arm is kept by the registered rule.

Arm	Cell kept	Bound	CV epoch	Mean CV Score	Fold Scores	Gap to baseline
SC (selected: AuscultTrust)	sc + static	static	14	**60.51**	52.2, 66.9, 62.3	+1.54
SC + random gain	sc_gain	dynamic	2	60.01	54.5, 65.2, 60.4	+1.04

Arm	Cell kept	Bound	CV epoch	Mean CV Score	Fold Scores	Gap to baseline
Baseline	baseline	-	2	58.97	52.9, 64.0, 60.0	0
Freq-MixStyle	freq_mixstyle	-	3	58.71	52.4, 61.3, 62.5	-0.26

Other cells: SC with the dynamic bound 58.44 (epoch 26); SC + random gain with the static bound 59.36 (epoch 1).

[Interpretation] The registered rule selects SC with the static bound. Three caveats: only three folds, with fold Scores spanning 52-67, so gaps of 1-1.5 points are inside the spread; the best CV epoch is 1-3 for several arms, which points to noise peaks; averaged over epochs 10-50 the paired gaps to the baseline are +1.3 (SC static), +1.7 (SC + random gain, static), -0.1 (SC dynamic), -0.2 (SC + random gain, dynamic) and +0.5 (Freq-MixStyle). So the static bound looks consistently useful and the dynamic bound and the random gain do not. The rule was not changed after seeing this.

5. Status and next steps

- **Done:** Part 1 for AST, HTS-AT, OPERA-CT, CLAP, HeAR (all cells, five seeds), DASS fine-tune (both methods), DASS frozen Patch-Mix CL; screening (18 jobs).
- **Running:** DASS frozen with Patch-Mix (4 of 5 seeds done); final refit of the four arms at their CV epochs, seeds 0-4 for all arms and 5-9 for the baseline and AuscultTrust (30 runs, test set scored once), expected to finish on the morning of Oct 9; this will give the non-inferiority check of AuscultTrust against the baseline.
- **Not started:** leave-one-device-out training, KAUH evaluation with frozen weights and the isolation contrast (needs code), SC arms for the other encoders.

6. Reproducibility

- Code: `auscult-trust/` (git, branch main); configs in `experiments/train/`; entry point `scripts/run.sh` (`reproduce`, `screen`, `summary`, `final`, `resume`).
- Results: `outputs/train/repro-wave1/` (Part 1; remote machine for HTS-AT, DASS fine-tune and HeAR cells, local for the rest), `outputs/train/screen-ast/` (screening and refit). Per-run `report.json` holds the full per-epoch history, per-device metrics and per-class recall.
- Decision log with the audit of the encoder inputs and the config check against the papers: `docs/decision_logs/2026-10-07-training-sweep.md`.
- Known deviations: the papers' exact seeds are not stated (we use 0-4); weighted cross-entropy is not used because every arm mixes; the frozen-encoder recipe (learning rate 1e-3) and Patch-Mix on non-AST encoders are our own design; DASS uses plain Patch-Mix without Lung-SRAD's blur.